



Low-cost foam cementitious electromagnetic metamaterial with bioinspired periodic array structure

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ABSTRACT

Developing cost-effective, weather-resistant, and broadband electromagnetic (EM) wave-absorbing building materials is critical for modern infrastructure. Herein, five bioinspired foam cement-based metamaterials (FCMs) are firstly designed, mimicking diverse biological structures ranging from the chameleon skin, moth eyes, proteinaceous spherules on the leafhopper cuticle, crocodile skin, and the *Apriona germari* wing hairs. Numerical simulations and experimental characterizations reveal that geometric modulation crucially dictates EM absorption performance. Notably, the FCM with a periodic pyramid array architecture bioinspired by the *Apriona germari* wing hairs achieves superior impedance matching and robustness. It exhibits a broad effective absorption bandwidth (−10 dB) spanning 3.36–18 GHz, maintaining high efficiency across wide incident angles (0°–60°) and polarization insensitivity. By synergizing the porous cementitious substrate with macroscopic metastructures, this design maximizes multiple loss mechanisms to overcome traditional bandwidth limitations. These findings establish bio-inspired FCMs as a scalable, high-performance solution for advanced EM protection in the prospective smart buildings.

1. Introduction

With the rapid development of advanced electronic devices, intelligent broadband technology and multi-band radar detection, ubiquitous electromagnetic (EM) radiation in the working and living environment generated by various electronic equipment has explosively increased, impairing human health, information security, electrical appliance operation, and military stealth applications [1–4]. As the most important place for daily human activities, constructions or buildings are the primary carriers to defend our health and normal life. However, the conventional architectural design and construction usually do not consider the protection of EM radiation in the living environment. Applying EM wave absorption materials covering the building surface is an efficient way to attenuate the EM energy and prevent the harmful effect of the radiation [5]. Thus, to eliminate the risks of EM pollution,

information leakage, and EM interference impacting the normal running of sophisticated electronic apparatus, the exploitation of novel and high-efficiency EM wave absorbing green building materials is urgently pursued in civil or military fields [6,7]. Among various building materials, cement exhibits a wide range of employment due to abundant resources, excellent environmental adaptability, good durability, simple preparation process and low-cost, which are still the most prevailing and economical basic building materials in civil engineering and military constructions. Herein, the development of cement-based building materials with high EM wave absorption and shielding functions arouses great interest and has become a hot topic in novel green building materials recently [5,8].

Due to the inherently weak EM absorption capacity of Portland cement itself, the EM absorption function of cement-based materials is normally achieved in two major approaches: filling with EM absorbents

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and the optimal structural design [2,8]. Specifically, for the first classification, various high EM loss-absorbing fillers such as carbon-based absorbents [5,9,10], magnetic absorbents [11,12], dielectric metal absorbents [13–15], ceramic absorbents [4,16], industrial by-products and solid wastes [15,17] are incorporated into the cement pastes. Nevertheless, blindly pursuing a large dissipation capacity will be converted into a contrary situation. Most of the incident EM waves are reflected directly on the surface of material due to the impedance mismatching between the bulk cement materials and free space. On this account, to balance the requirements of impedance matching and high dissipating ability, reasonably structured cement-based composites are fabricated to advance the EM wave absorbing performance. Diverse structural formats such as porous [4,18], multi-layer [19,20], 3D cross-link foam [21], and periodic array structures [9,14,22,23] are exploited to obtain higher absorbing strength and wider broadband. However, the EM wave absorption of cement-based materials in the low-frequency range (1–6 GHz) and the varying incident angles are far from satisfactory. Therefore, an absorber with ideal EM wave absorption performance is still craved, which can perfectly coordinate the contradiction between impedance matching and high EM exhausting intensity.

Metamaterial absorbers are innovative EM wave absorbing materials that possess good impedance matching, high-efficiency wave absorption, and resonant frequency selection through structural design [24–26]. Recently, broadband metamaterial absorbers with multi-layer

structures have attracted great attention, which achieve a coupling effect owing to the various resonant frequencies or optimized impedance matching characteristics. For example, Wang et al. [27] proposed a cement-based metamaterial absorber consisting of CNT conductive gratings and cement composite dielectric layers, achieving a broad effective EM absorption bandwidth of 13 GHz (reflection loss (RL) < –15 dB) within the 1–18 GHz range by optimizing structural parameters and impedance matching. However, several limitations remained, such as a limited extensible frequency range (especially in the low-frequency band), high density, and a complicated structure. Alternatively, researchers have studied 3D metamaterials combined with strong absorbents and found that even single-layer metamaterials can have excellent EM absorbing performance. For instance, Sun et al. [26] developed 3D reduced graphene oxide (rGO) sponge metamaterials featuring frustum pyramid units, which exhibited favorable impedance matching and an ultra-wide effective absorption bandwidth (RL < –10 dB) across the 2–40 GHz frequency range. However, the preparation process of such a nanomaterial-based sponge is rather complex and demanding, costly, poisonous, and not environmentally friendly, which will seriously restrict its application in practical construction engineering and be far from the green building concept. Recently, researchers have drawn great inspiration from structural and functional characteristics of biomaterials, developing high-performance of bionic industry products and architectures (as shown in Fig. 1) [28–34]. Based on above insights,

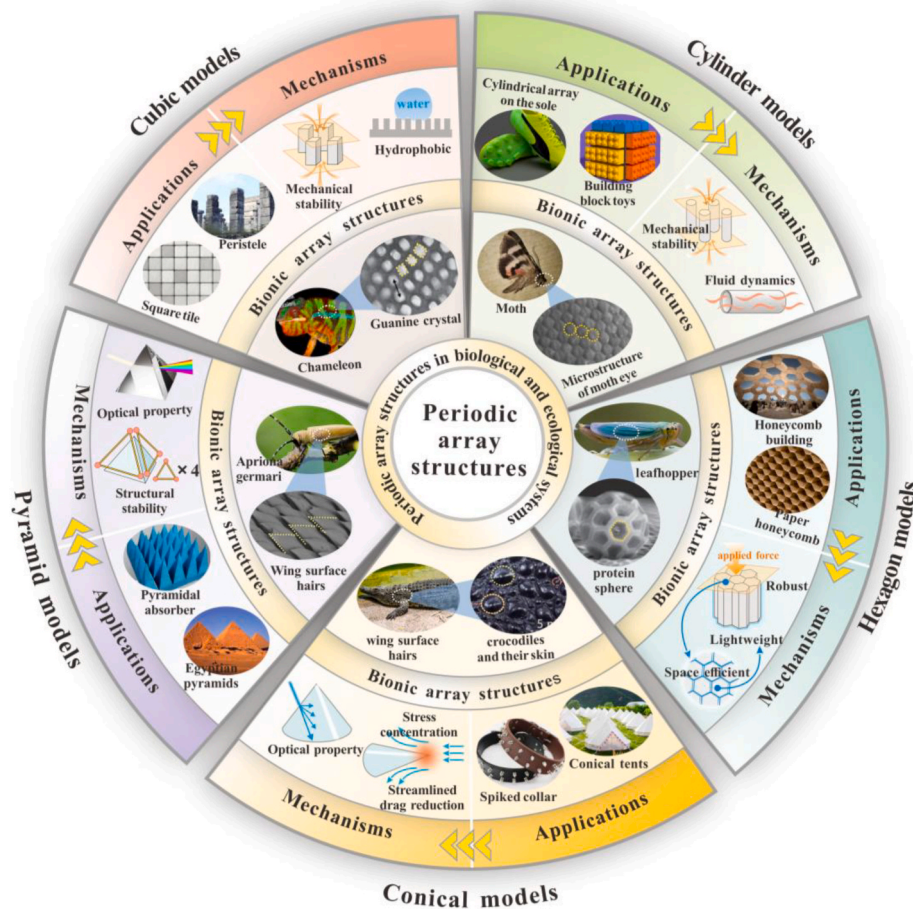


Fig. 1. Periodic array structures in biological and ecological systems, as well as their biomimetic application and mechanisms in daily life: The arrangement of guanine crystals in the form of cubic arrays on the chameleon's skin. The chameleon photo and SEM image of guanine crystals are from licensed reproduction [28]. Copyright 2017, ACS. The cylindrical array arrangement structure on the eyes of a moth. The moth picture and SEM image of moth's eyes are from licensed reproduction [29]. Copyright 2023, Elsevier Ltd. The hexagonal hollow protein spheres covered the surface of leafhopper. The SEM image of protein spheres on the skin of leafhoppers is from licensed reproduction [30]. Copyright 2013, RS. The conical array structure on the crocodile skin. The microstructure image of crocodile skin is licensed for reproduce [31]. Copyright 2025, Wiley-VCH. The periodic pyramid structure that appears on the wings of the *Apriona germari*. The SEM image of *Apriona germari*'s wing is from licensed reproduction [32]. Copyright 2025, Wiley-VCH.

Wang et al. [5], fabricated a bionic groove-shaped all-cementitious-dielectric metamaterial for EM wave absorption that was inspired by the microscopic features on butterfly wings. As a result, an ultra-broadband effective absorption of 34.6 GHz (in 1–40 GHz) over 0° to 60° incident angles was attained. Nevertheless, the directional groove design of this material exhibited significant polarization-dependent absorption performance, coupled with a high density, which would limit its protection application under a complex EM environment. On the other side, EM wave absorbing metamaterials featuring periodic biological structures such as cubic [14,26], cylinder [22], hexagon [18,26,35], conical [36,37], and pyramids [38] have been proved to activate diverse interference, diffraction, and scattering mechanisms at varying incident angles, exhibiting exceptional physical properties in optics and electromagnetism. However, former bioinspired absorbers are fabricated from resin, aerogel, or other polymer matrices [4,14,39], and no existing research

has integrated such bioinspired metasurface designs with porous cementitious building materials. This combination of the metasurface and the cementitious matrix not only facilitates the development of ultra broadband and high attenuation microwave absorbers but also fulfills the requirements for lightweight properties, fire resistance, and environmental friendliness in high performance green building materials. Nonetheless, it still remains unclear which kind of biomimetic structural type is more suitable for combination with the foam cement-based absorbing matrix, as well as the relationship between the structural design of metamaterials and the EM absorption performance.

This work is dedicated to exploring the structural dependence of biomimetic periodic array structural cement-based foam metamaterial (FCM) at a macroscopic scale on EM loss performance. A simple constitution and preparation method for the FCM is proposed, using multiwalled carbon nanotubes (MWCNTs) as the absorbent, and

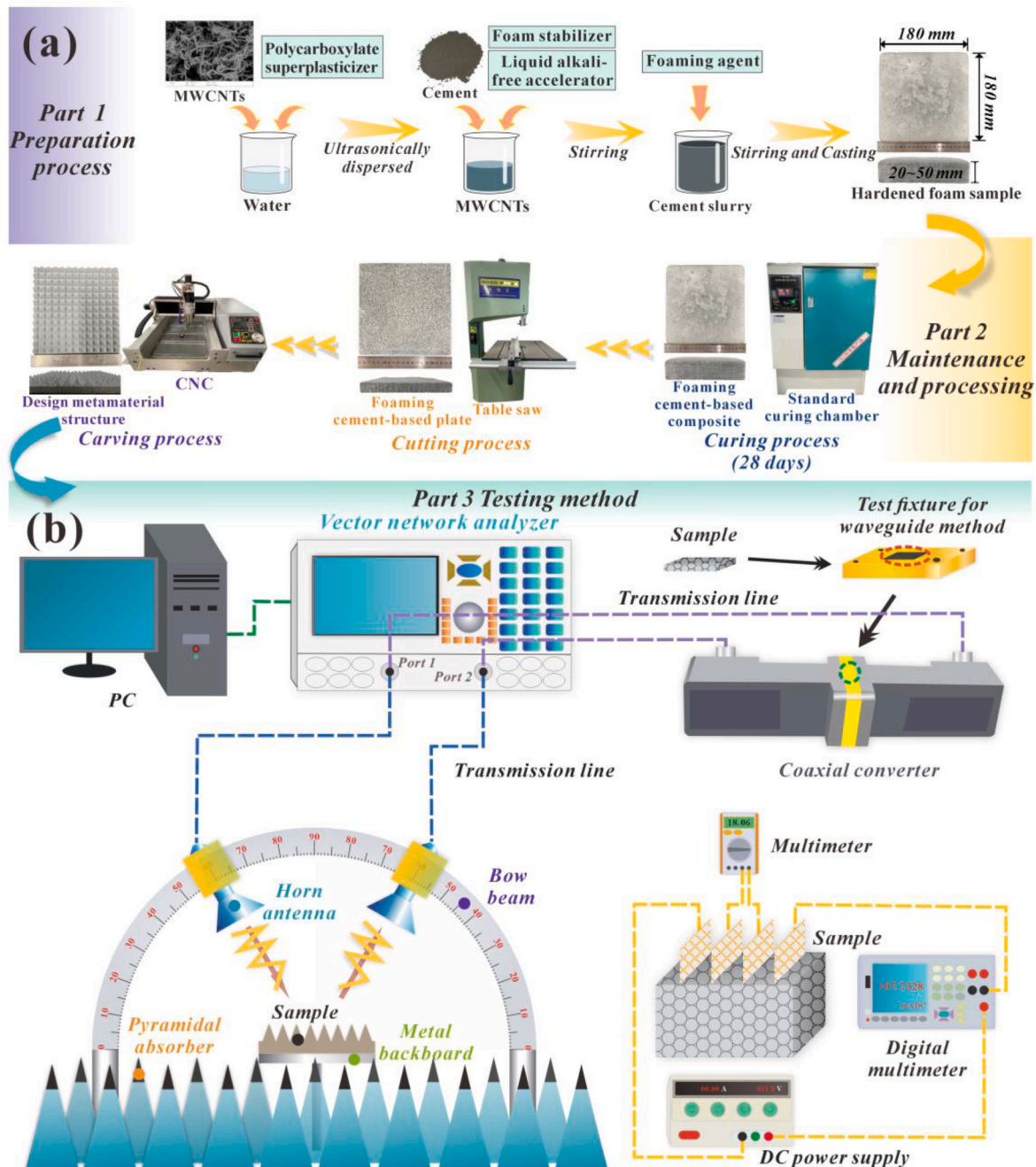


Fig. 2. The preparation process and testing methods of FCM. (a) The preparation process of FCM; (b) the EM performances testing methods.

applying foam cement composites as the matrix. Inspired by the chameleon skin, moth eyes, proteinaceous spherules on the leafhopper cuticle, crocodile skin, and the *Apriona germari* wing hairs, the array structured FCM in different biological configurations, such as cubic prism (FCM-cp), cylinder (FCM-cy), hexagonal prism (FCM-hp), conical (FCM-c) and pyramid (FCM-p) is explored using the simulation software. The results indicate that the EM absorbing effect of FCM metamaterials are seriously influenced by their geometry shapes and parameters. Among them, the FCM-p demonstrates the optimal absorption effect and is further manufactured by a computer numerical control carving machine (CNC). The superior microwave absorption of FCM-p stems from its unique structural features unavailable in other biomimetic configurations. FCM-p allows incident EM waves with varying angles and polarizations to enter the inter-pyramid gaps, where strong resonance occurs to enable full EM energy dissipation. In addition, its continuous gradient impedance along the incident direction minimizes interfacial reflection, the diffraction from its vertices and edges extends wave propagation paths and enhances energy dissipation, its four-fold rotational symmetry further delivers polarization insensitivity and highly robust absorption at wide incident angles. Ultimately, the lightweight FCM-p offers desirable impedance matching and EM loss, which has ultra-broadband (3.36–18 GHz) and a wide incident angle range (0°–60°) effective absorption in all polarization modes, which shows an impressive robustness.

2. Experimental programs

2.1. Raw materials and sample preparation procedure

The raw material information for preparing FCM was listed in Text S1, and the preparation process and mix proportion of the foam cement-based composites infilled with MWCNTs were elaborated in Fig. 2(a) and Table S2 [21], respectively. Firstly, the MWCNTs were pre-mixed with the polycarboxylate superplasticizer in the pre-weighted water and ultrasonically dispersed for 30 min to acquire the MWCNTs dispersion. This combined ultrasonic and stirring treatment effectively eliminated MWCNTs agglomeration and guaranteed uniform dispersion in the cement matrix. Then, the pre-weighted cement and foam stabilizer were input into the MWCNTs dispersion and mixed for 1 min (1000 r min⁻¹) to obtain the fresh cement paste. After that, the foaming agent was added into the cement paste at once and rapidly stirred for 10 s (6000 r min⁻¹) to obtain a universal gas distribution. Then, the well-dispersed cement paste was poured into plastic molds (180 mm × 180 mm × *d*, and *d* = 20–50 mm) for the foaming standing process. Furthermore, after the initial curing period of 24 h, the hardened foam samples were demolded and transported into a standard curing chamber (20 ± 1 °C and relative humidity ≥ 90%) to continue curing until 28 days, and then a hardened foaming cement-based plate was obtained. A table saw was used to cut the irregular part of the plate. Eventually, the CNC was employed to process the FCM plate based on the design metamaterial structure.

Notably, the compressive strength and flexural strength values of the foamed cementitious matrix at the designated density (0.90 ± 0.05 g cm⁻³) were tested as 13.37 MPa and of 2.61 MPa, respectively, as detailed in our published study [21], which identified a good mechanical performance of the foam cementitious matrix. Based on above findings, this work focused on the exploration of macroscopic bio-inspired structural designs, preparation and the synergistic EM absorption mechanisms. All samples were prepared with the actual density controlled at 0.90 ± 0.05 g cm⁻³.

2.2. Characterization and instruments

As depicted in Fig. 2(b), the complex permittivity ϵ and permeability μ of the foam cement-based composite were tested in the frequency of the X band (8–12 GHz) using a vector network analyzer (CLP Kesi

Technology Co., Ltd, AV3629D, China) by the waveguide method according to the Chinese standard GB/T 35679-2017. Furthermore, the EM wave absorbing performance of the FCM was analyzed by an arched reflection testing method employing a vector network analyzer system (VNA, Keysight E5071C, U.S.) covering the 2–18 GHz band. The electrical conductivity of the foam cement-based composite incorporated with MWCNTs was measured by a four-electrode voltammetry method by a digital multimeter (KEYSIGHT 34460A), and an adjustable DC regulated power source (MAISHENG MD15010D). The electrical conductivity (σ) of the sample was obtained by formula (1) [14]:

$$\sigma = \frac{IL}{US} \quad (1)$$

Specifically, U , I , S , and L were the testing sample's voltage, current, cross-sectional area, and the distance between two inner electrodes, and the average value of three independent samples was applied to eliminate experimental error.

Moreover, the morphology and microstructure of foam cement-based composites were examined by scanning electron microscope (SEM) (Carl Zeiss Micrographic Image Co., Ltd, Germany, EVO MA15).

2.3. Structural design descriptions

As exhibited in Fig. 3, inspired by different biological structures of the chameleon skin, moth eyes, proteinaceous spherules on the leafhopper cuticle, crocodile skin, and the *Apriona germari* wing hairs, five geometrically distinct periodic-array FCM absorbers (FCM-cp, FCM-cy, FCM-hp, FCM-c, and FCM-p) were designed accordingly. Besides, the geometrical simulation parameters of the different array units for FCM absorbers were illustrated in Table S3. A three-dimensional EM high-frequency structure simulator (HFSS) software was used to analyze the microwave absorbing performance, such as reflectivity, electric field distribution, power loss intensity, and current density distribution of the FCM through the finite element method [26]. Herein, an elementary unit cell was established to simulate the arch reflection method test in the 2–18 GHz frequency range, where a Floquet port with periodic boundary conditions was assigned [40], and a perfect electric conductor (PEC) was set at the bottom of the sample. The microwave absorption performance was mainly characterized by RL (Eq. (1)), which was determined by the input impedance Z_{in} of the absorber and the free space impedance Z_0 (equal to 377 Ω), while Z_{in} was correlated with the intrinsic EM parameters of the material, as shown in Eq. (2). Where f presented the frequency, c presented the speed of light in vacuum, and d was the thickness of the material. The EM wave incident angle was varied from 0° to 80° in TE and TM modes. Simultaneously, the EM parameters of foam cement-based composites and the structural parameters of the model were designated in the HFSS software.

$$RL = 20 \log \left| \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \right| \quad (2)$$

$$Z_{in} = Z_0 \sqrt{\frac{\mu_r}{\epsilon_r}} \tanh \left(j \frac{2\pi f d}{c} \sqrt{\mu_r \epsilon_r} \right) \quad (3)$$

3. Results and discussions

3.1. The basic property of the foam cement-based composites

The EM loss capacity of EM wave absorbing materials was directly reflected by the complex permittivity ($\epsilon_r = \epsilon' - j\epsilon''$) and complex permeability ($\mu_r = \mu' - j\mu''$), which were obtained through the waveguide method. The dielectric loss tangent angle ($\tan \delta_\epsilon = \frac{\epsilon''}{\epsilon'}$) and the magnetic loss tangent angle ($\tan \delta_\mu = \frac{\mu''}{\mu'}$) were used to calculate the EM loss capacity of EM wave absorbing materials, as well [41,42]. As presented in Fig. 4(a, b) attributing to the MWCNTs as a strong dielectric

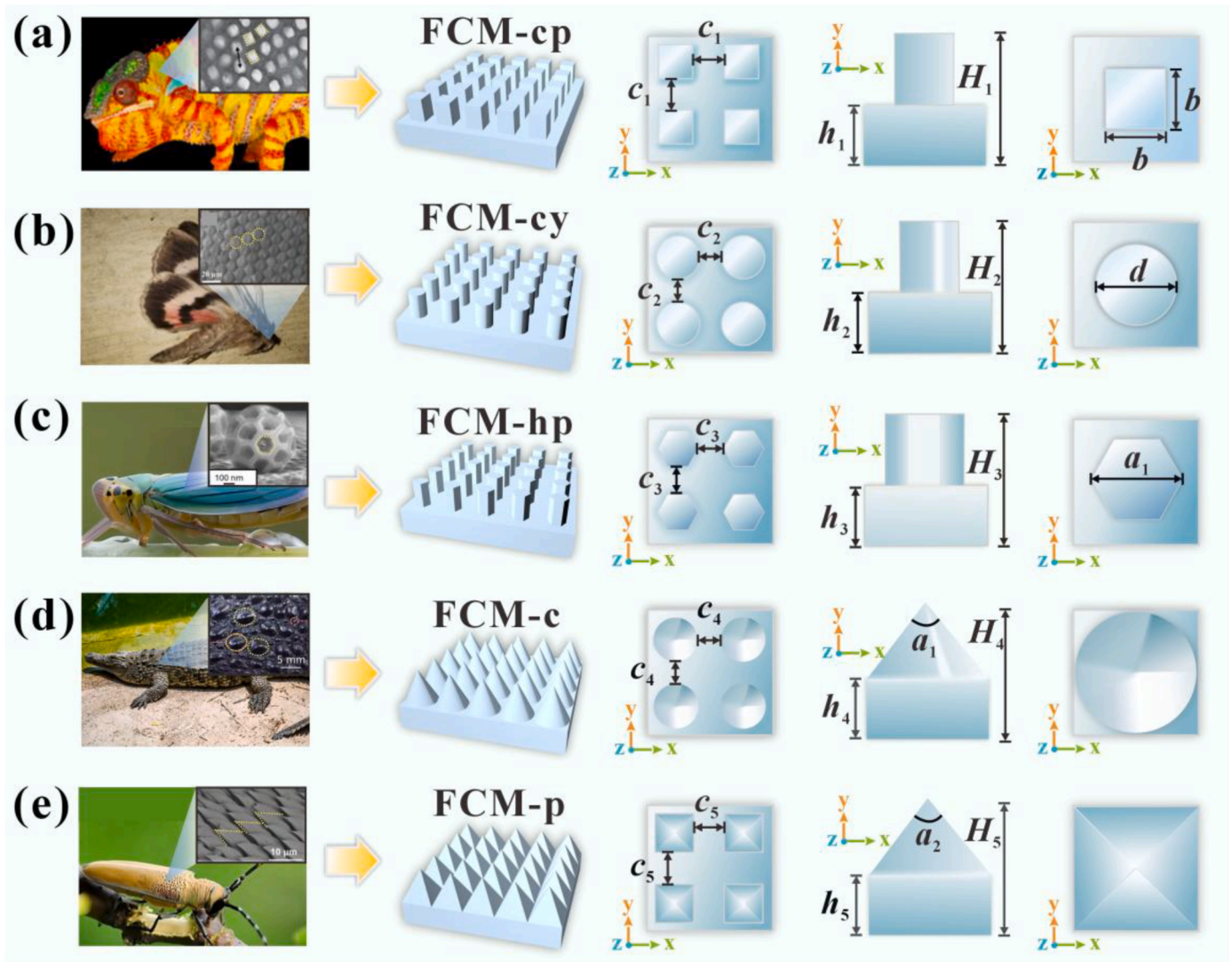


Fig. 3. Design of different periodic array structure types of FCM metamaterials inspired by biology. (a) Chameleon skin. The chameleon photo and SEM image of guanine crystals are from licensed reproduction [28]. Copyright 2017, ACS. (b) Moth eyes. The moth picture and SEM image of moth's eyes are from licensed reproduction [29]. Copyright 2023, Elsevier Ltd. (c) Proteinaceous spherules on the leafhopper cuticle. The SEM image of protein is from licensed reproduction [30]. Copyright 2013, RS. (d) Crocodile skin. The microstructure image is from licensed reproduction [31]. Copyright 2025, Wiley-VCH. (e) The *Apriona germari* wing hairs. The SEM image of *Apriona germari* wing hairs is from licensed reproduction [32]. Copyright 2025, Wiley-VCH.

material without magnetic loss properties, the real permeability of FCM was close to 1, and the imaginary part of permeability was almost 0. On the other side, the real permittivity value of FCM was about 3.7, while the imaginary part was approximately 0.53, indicating a strong dielectric loss capacity. Meanwhile, the FCM also has a distinct dielectric loss tangent angle of 0.15, but the magnetic loss tangent angle of FCM was 0, further illustrating that only dielectric loss existed in the FCM. Notably, due to the intrinsic properties of the composite was non-magnetic, the minor fluctuations in permeability observed in the test results were originated from systematic errors in the waveguide testing method, including system background noise and minute assembly gaps between the sample and the fixture. These minor fluctuations almost did no contribution to electromagnetic wave attenuation, as the material's absorption performance was entirely dominated by dielectric loss (see Fig. 5).

Due to the air filling in the homogeneous porous skeleton structure of the foam cement, FCM was lightweight with a density below 0.9 g cm^{-3} , which was fascinating for building walls, fireproofing, vibration absorption, filling materials, and heating insulation applications. The micro fibrous MWCNTs were well dispersed and interconnected in the

pore wall of the foam composite, as exhibited in Fig. 4(c and d). Additionally, electrical conductivity results demonstrated that the electrical resistivity of the sample filled with MWCNTs was $4.9 \times 10^{-4} \text{ S m}^{-1}$, which was as high as 49 times of the blank sample ($1.0 \times 10^{-5} \text{ S m}^{-1}$) ascribed to forming a three-dimensional conductivity network cellular structure (as shown in Fig. 4(c)). Moreover, with a higher conductivity, the induced current occurred in the FCM under the excitation of the microwave so that the EM energy was converted into electric energy, and resistance loss would be generated. The high-magnification SEM image analysis was provided in Text S2.

Therefore, the dielectric and conductive loss mechanisms have synchronously taken place in the FCM matrix, and the polarization loss (ϵ_p'') and conductive loss (ϵ_c'') were deduced from ϵ'' based on the Debye theory. Hence the ϵ_p'' and ϵ_c'' were calculated by formula (4) and (5) [43, 44]:

$$\epsilon_c'' = \frac{\sigma}{\omega \epsilon_0} \quad (4)$$

$$\epsilon_p'' = \frac{\epsilon_s - \epsilon_\infty}{1 + \omega^2 \tau^2} \omega \tau = \epsilon'' - \epsilon_c'' \quad (5)$$

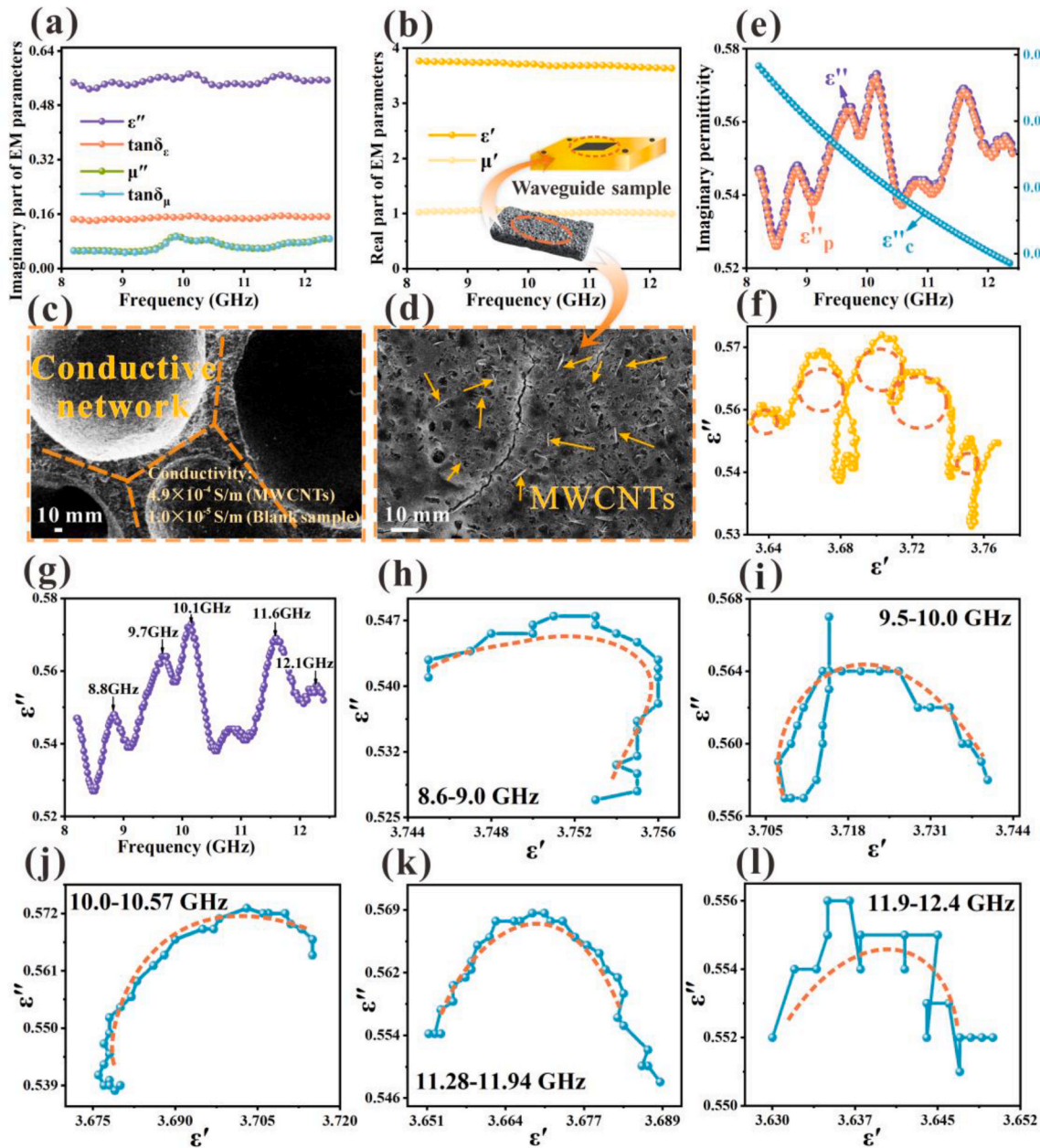


Fig. 4. The EM properties and microscopic morphology of FCM. (a-b) The EM parameters of FCM; (c-d) the SEM image of MWCNTs inside the FCM (MWCNTs were indicated by the yellow arrows); (e-l) the Cole-Cole semicircle of FCM. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Here, ω was the angular frequency, σ was the electrical conductivity, ϵ_0 was the permittivity of vacuum, ϵ_s was the static permittivity, ϵ_∞ was the dielectric constant at infinite frequency, and τ was the relaxation time.

Basic dielectric relaxation mechanism was described in Text S3 [21, 45]. As shown in Fig. 4(e), to further explain the influence of FCM on wave-absorbing performance, the Cole-Cole semicircle corresponded to the 8-12 GHz frequency band of the sample. From Fig. 4(f), five Debye relaxation processes could be found in the relatively narrow range of 8.8 GHz, 9.7 GHz, 10.1 GHz, 11.6 GHz, and 12.1 GHz, respectively. The above Debye relaxation effects were represented by the real and imaginary parts of the dielectric constant, as shown in Fig. 4(g-l) (Fig. 4(h)-(l) were zoomed-in views of representative local regions to clearly demonstrated individual relaxation semicircles). These multiple polarization processes were attributed to the high heterogeneity of the

composite, specifically stemming from defect polarization within the functionalized MWCNTs, interfacial polarization between the MWCNTs and cement hydration products such as C-S-H gel, as well as macroscopic polarization at the abundant air-pore and solid matrix interfaces. Therefore, the foam cement matrix filled with MWCNTs exhibited superior dielectric loss capacity.

3.2. The EM wave absorbing performance of FCM with different bionic array structures

The absorbing performances of the five bionic structures of FCM with periodic arrangement were compared in Fig. S2-Fig. S6. The RL in the frequency range of 2-18 GHz was simulated using the HFSS software, which demonstrated that the EM wave absorbing property of metamaterials with diverse bionic unit shapes and geometry parameters were visibly different. Thereinto, when the total height H was set as 20 mm,

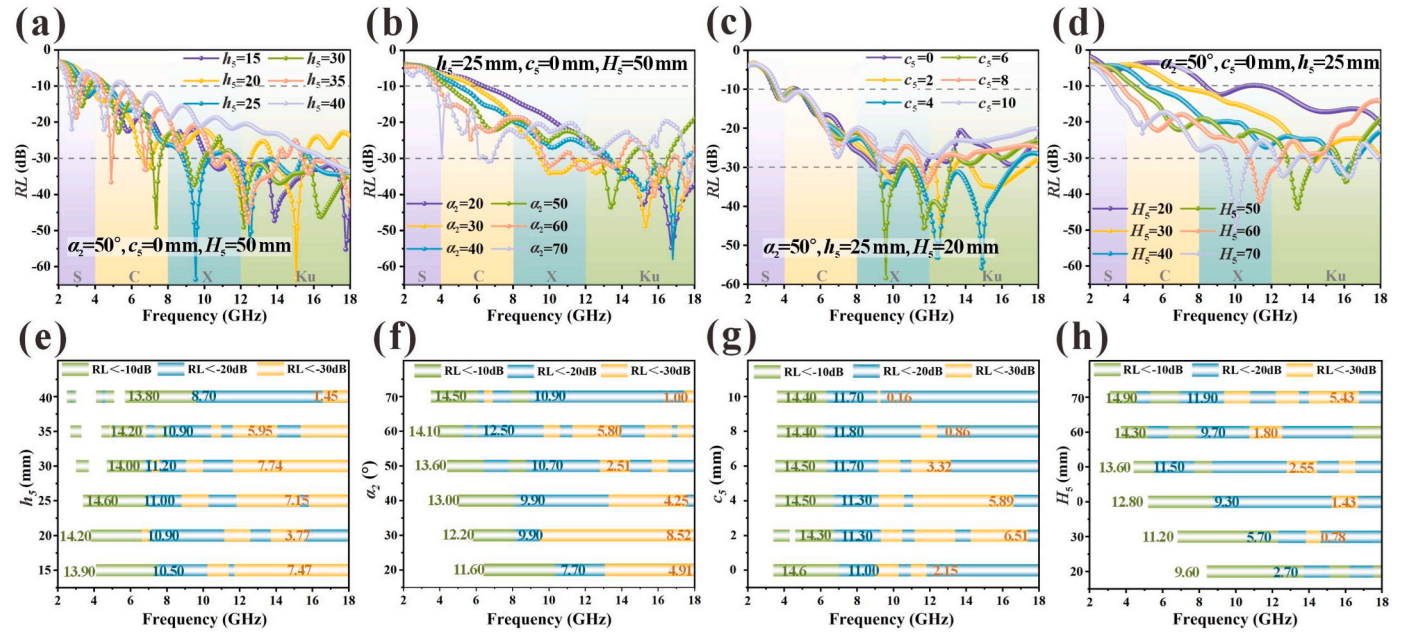


Fig. 5. The simulation reflectivity of FCM-p metamaterials with a total height of 50 mm under varying geometry parameters. (a) h_5 ; (b) α_2 ; (c) c_5 ; (d) H_5 ; (e-h) the corresponding bandwidth for RL less than -10 dB, -20 dB, -30 dB.

the RL curves of the FCM-cp, FCM-cy, and FCM-hp exhibited a discontinuous effective absorbing frequency range in the S, C, and X bands, which were difficult to reach -10 dB indicating relatively weak EM wave absorbing performances. Alternatively, FCM-p and FCM-c showed significantly better EM wave absorbing performance than other aforementioned metamaterial structures, the effective absorption band below -10 dB was continuous and almost covered 8–18 GHz, which indicated that 90% of EM wave energy has been dissipated. Furthermore, the effective absorption band was well enhanced with increasing sample height. Specifically, as illustrated in Fig. S7 and Fig. 5, FCM-p exhibited an optimal EM wave absorbing performance when the sample height exceeded 50 mm, effectively absorbing band below -20 dB within the frequency range of 4–18 GHz and dissipating up to 99% of EM energy. Therefore, the results declared that the geometry designs and their macrostructure parameters of FCM metamaterials both acted as crucial elements in EM wave-absorbing properties. Notably, while a 50 mm thickness was substantial compared to conventional coatings, the ultra low density of the FCM-p (<0.95 g cm $^{-3}$) ensured its direct application as structural functional integrated prefabricated wall blocks or decorative claddings. This provided superior EM protection without imposing a significant dead load on the building structures.

Furthermore, the existing literature about cementitious EM wave absorber materials was primarily focused on a perpendicular incidence. However, in actual scenarios of EM wave absorption, the EM waves would come from different directions in various polarization modes. The EM absorption performance of FCM-p under various incident angles and polarization modes was shown in the contour plots of Fig. 6(a and b), where the optimized structural dimensions were fixed as $H_5 = 50$ mm, $h_5 = 25$ mm, $\alpha_2 = 50^\circ$, and $c_5 = 0$ mm. Across both TE and TM polarization modes, the FCM-p achieved an effective bandwidth ($RL < -10$ dB) of 6–18 GHz and a strong absorption bandwidth ($RL < -20$ dB) of 8–18 GHz over incident angular ranges of 0–60° and 0–50°, respectively. Notably, at high frequencies (>9 GHz), over 90% absorption was retained even at 70° incident angles. Above results have proved the material's remarkable adaptability to different incident angles and polarization modes. Moreover, the electric field and Poynting vector distribution of the FCM-p under TE and TM polarization modes with varying polarization angles at the absorption peak of 9.52 GHz were further illustrated in Fig. 6(c and d). When EM waves were incident

perpendicularly ($\theta = 0^\circ$), the electric fields in both TE and TM modes align horizontally. The multiple reflections and refractions of EM wave resulted in a significant electric field enhancement zone formed between adjacent pyramid units under TM mode, causing the Poynting vector to enters the material from the pyramid sides. However, under the TE mode, a distinct energy entry path emerged in the FCM-p, with the Poynting vector mainly penetrating from the pyramid apex. As the incident angle increased, the EM wave gradually deviated from the Z-axis, deflecting toward the X direction and causing a corresponding tilt in both the electric field and Poynting vectors. When the incident angle increased to 60°, the wave-facing side pyramid facet became the dominant entry point for energy. Meanwhile, a fraction of the incident wave that reached the gaps between pyramidal units was reflected to the side opposite to the incident side, which led to a localized enhancement of the electric field. In contrast, at an extreme wide angle of $\theta = 80^\circ$, a significant energy dissipation only occurred at the pyramid apex due to strong diffraction effects induced by its sharp geometry. Although the EM wave propagated almost horizontally at such a large angle, the Poynting vectors distribution revealed that EM energy could still be delivered deep into the material via diffraction and other interactions, confirming that the good adaptability of FCM-p to wide incident angles and diverse polarization modes.

3.3. Comparison between the testing results and simulated calculation

Based on the above discussions regarding the optimized design from different bionic structures, FCM-p metamaterial absorbers with total thicknesses of 20 mm and 50 mm were prepared separately, as illustrated in Fig. 7(a). When the total thickness of the sample was 20 mm, the FCM-p exhibited an $RL < -10$ dB across the frequency range of 7.92–18 GHz (encompassing the entire X and Ku bands), with a pronounced absorption peak of -22.63 dB observed at 14.32 GHz, as displayed in Fig. 7(b). Furthermore, as the sample thickness increased to 50 mm, the interference peaks of FCM shifted toward lower frequencies, which was caused by the quarter-wavelength effect, as shown in Eq. (6) [11,46,47]. Meanwhile, the absorption band of FCM-p broadened significantly, achieving over 90% absorption across the frequency range of 3.36–18 GHz (encompassing the entire C, X, and Ku bands). On the other side, the absorption band for $RL < -20$ dB was also dramatically expanded,

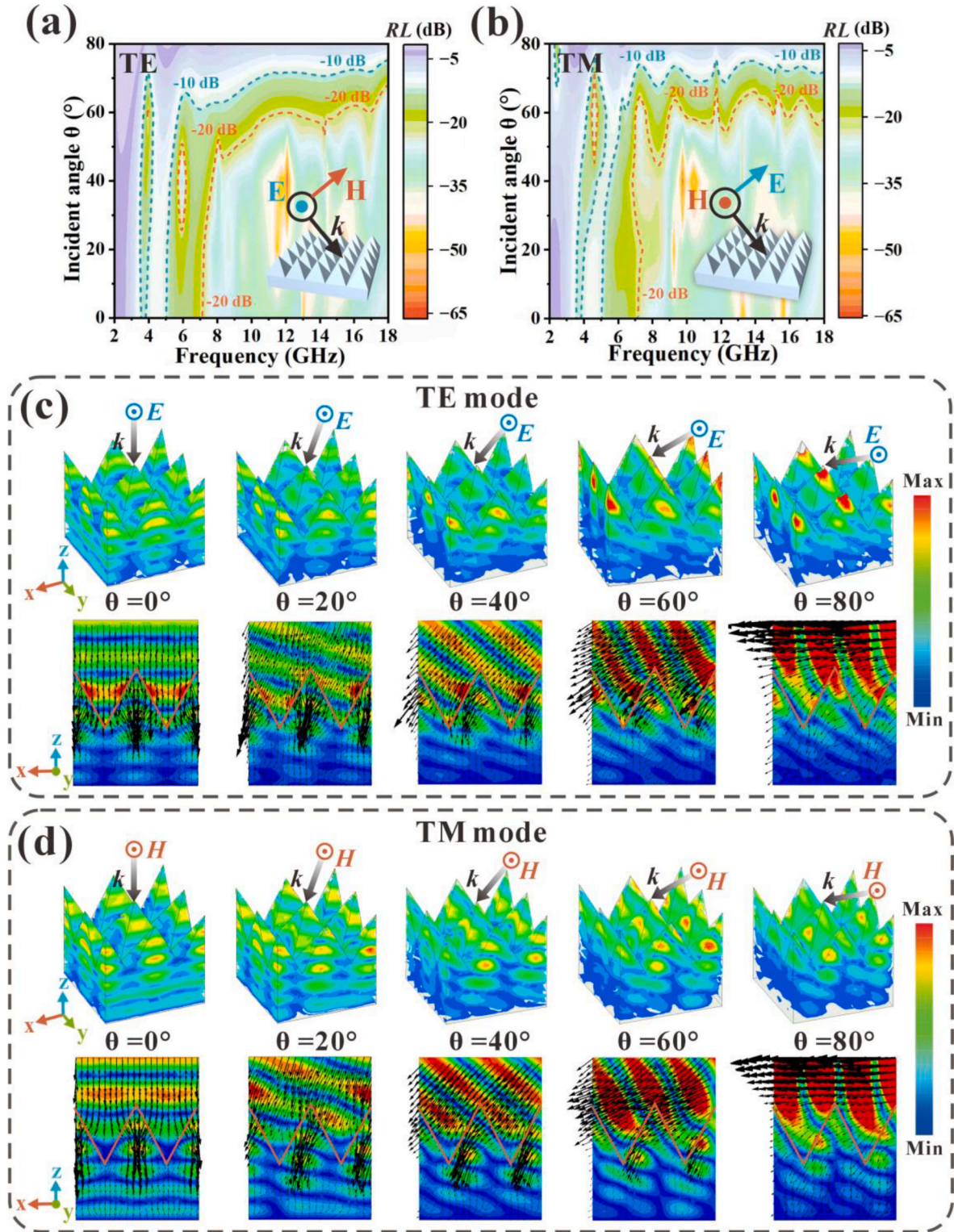


Fig. 6. Angular and polarization stability characteristics of FCM-p. (a) The RL contour maps of FCM-p in TE mode and (b) TM mode. (c) The simulated electric field and Poynting vector distributions in TE mode and (d) TM mode under oblique incidence.

covering the range from 7.04 to 18 GHz, with a minimum RL value of -63.53 dB observed at 9.52 GHz. Notably, the measured RL curves of FCM-p matched the calculation results well, adequately confirming that FCM-p has exhibited outstanding EM wave absorbing performance. Nevertheless, considering the machining precision, experimental operational errors, and micro-structural differences in the FCM-p specimen, some minor discrepancies between the simulation and actual testing

results were still observed.

$$f_m = \frac{c}{4d\sqrt{\mu_r \epsilon_r}} (2n + 1), n = 0, 1, 2, \dots \quad (6)$$

Moreover, according to the radar cross section (RCS) formula [36, 47], the RCS performance of the FCM-p was also evaluated for its military potential via HFSS simulation at 9.52 GHz, comparing it to a

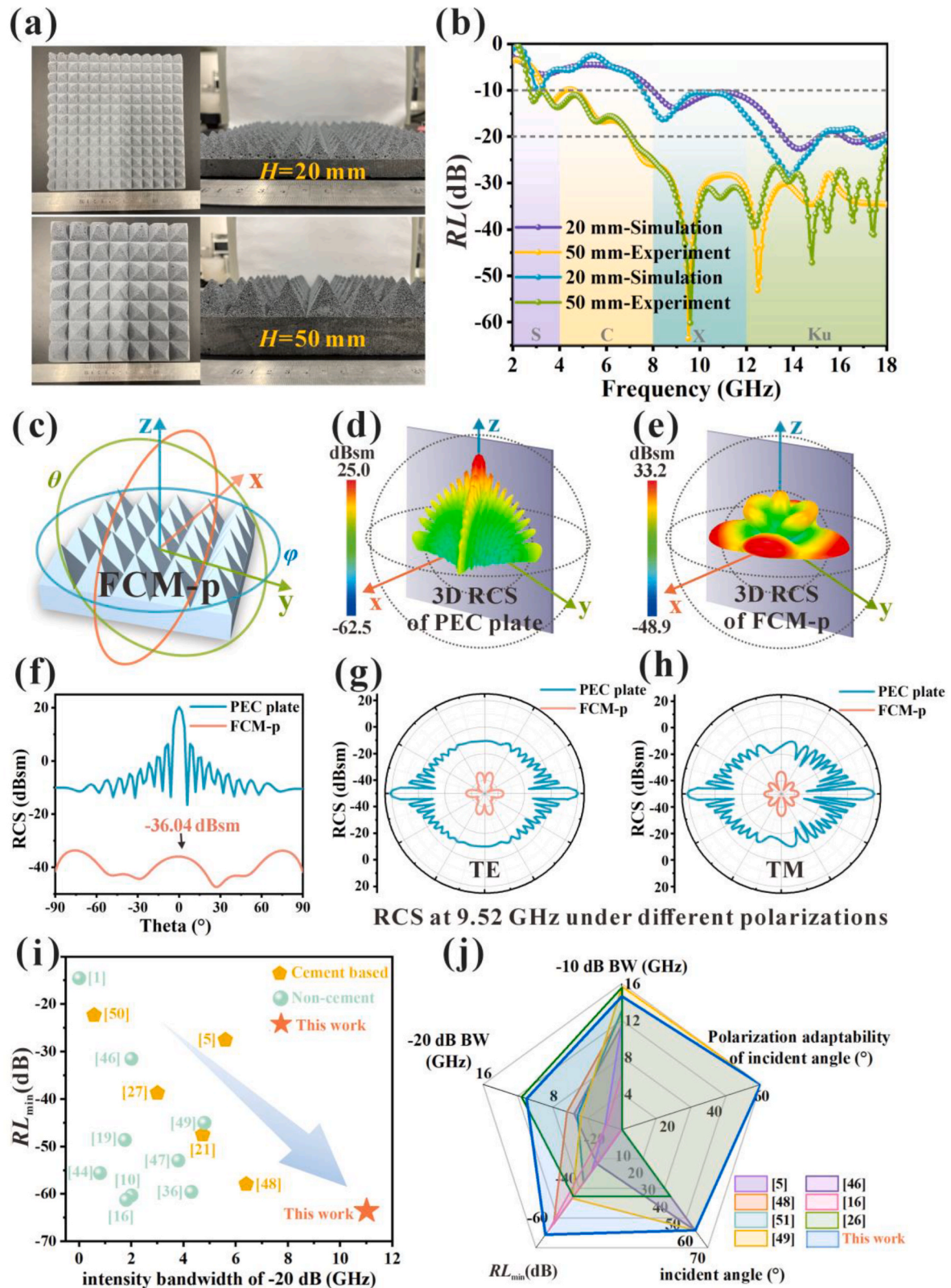


Fig. 7. Experimental verification and RCS reduction characteristics of FCM-p. (a) Photographs of the fabricated FCM-p samples with thickness of 20 mm and 50 mm. (b) Comparison between the measured and simulated RL curves. (c) The simulation model of FCM-p. 3D RCS signal of (d) the PEC plate and (e) FCM-p. (f) Comparison of RCS curves in Cartesian coordinates. Polar RCS scattering patterns in (g) TE mode and (h) TM mode. (i, j) Comprehensive performance comparison between FCM-p and other reported absorbers.

standard PEC plate (300 mm $\times$ 300 mm $\times$ 5 mm). The FCM-p model and its 3D RCS signal were separately depicted in Fig. 7(c–e). Here, φ and θ represented the azimuth and elevation angles, respectively. The 3D RCS signals have confirmed that the PEC plate produced sharp peaks across θ from -90° to 90° , whereas the FCM-p provided a markedly smoother RCS outline due to effective wave dissipation. Therefore, the Cartesian coordinate indicated an outstanding RCS reduction of the FCM-p, which reached -36.04 dBsm at $\theta = 0^\circ$ and -47.47 dBsm at $\theta = 90^\circ$, as quantified in Fig. 7(f). Additionally, compared to the PEC plate, the FCM-p sample achieved a consistently low monostatic RCS around -40 dBsm across the entire sweep ($\pm 180^\circ$) for both TE and TM modes, as shown in the RCS polar curve in Fig. 7(g and h).

Moreover, the comparison of absorption bandwidths and $RL_{\min}$ between several published EM wave absorbers [1,5,10,16,19,21,27,36,44,46–50] and the FCM-p metamaterial proposed in this work were displayed in Fig. 7(i). The results showed that FCM-p exhibited superior strong loss performance over most existing counterparts reported in the literature across an ultra-wide frequency band, achieving a strong loss bandwidth of $RL < 20$ dB as high as 11 GHz. A comparison between FCM-p and other existing EM wave absorbing materials regarding their adaptability in complex EM environments was presented in Fig. 7(j) [5,16,26,46,48,49,51]. It was evident that current researches on adaptability to incident angles and polarization modes of EM wave remained relatively insufficient. Overall, the comparative analysis has clearly demonstrated that the FCM-p proposed in this work exhibited comprehensive advantages across multiple aspects.

3.4. The EM wave absorption mechanism of the FCM

3.4.1. The distribution analysis of volume current density, electric field intensity and power loss

To have a better elucidation of the intrinsic EM wave absorption mechanism of the FCM, the distributions of volume current density, electric field intensity, and power loss intensity at the absorption peaks of various unit structures were comprehensively analyzed by HFSS simulation software, with the total thickness set to $H = 20$ mm. Based on the value of EM parameters about foam cement-based composite with their frequency-dependent characteristics, the current density distributions and vectors of FCM-cp, FCM-cy, FCM-hp, FCM-c, and FCM-p at their wave absorption peaks were evaluated in Fig. S8, respectively. At these frequency points, the absorber could be treated as a dielectric waveguide to analyze the propagation characteristics of EM waves within the intrinsic structure of metamaterial. As shown in Fig. S8(a–c), absorbing peaks in the mid- and low-frequency range of FCM-cp, FCM-cy, and FCM-hp occurred around 6.69–6.80 GHz, and the volume current density was primary concentrated at the interface between the upper array unit and basal layer. In parallel with this, as the frequency increased to 14.48–15.52 GHz, a substantial volume current density was observed predominantly on the array unit surfaces of the three distinct meta-structure types. But an intense current distribution was also evident at the interface between the cylinder and the basal layer of the FCM-cy structure, which was attributing to the diffraction phenomenon during the EM wave propagating process. Thus, different from, the FCM-cp and FCM-hp, when EM waves struck the lateral edge of the upper cylinder of the FCM-cy, the frequency band in which coherent loss occurred was shifted. Additionally, the diffraction effect gave rise to creeping waves that propagated along the surface, and a rather complex propagation path and the phase difference variations of the EM waves occurred.

In contrast, within a lower frequency range of 3.28–3.44 GHz, the current was predominantly horizontally aligned in the transition regions between the conical or pyramidal upper array structures and their substrates, as presented in Fig. S8(d and e). As the frequency increased, the current extended progressively along the central regions toward the upper and peripheral sections of the conical and pyramidal structures of the FCM-c and FCM-p, exhibiting a clearly swirling pattern. Meanwhile,

either current vector in the FCM-cp and FCM-hp was predominantly horizontally aligned, with only minor fluctuations was observed. Conversely, in the FCM-cy, FCM-c, and FCM-p samples, a clear vortex-like current distribution emerged, which was inherently relating to the rotational symmetry of the unit structure. Specifically, unlike the other three configurations, the FCM-c and FCM-p sample have exhibited gradient arrangements that associated with the direction of EM wave incidence, thereby demonstrating more pronounced phase deflection effects. When the upper geometry structure displayed a gradient variation along the Z-axis, the phase delay of incident EM waves occurred both on the top and side surfaces along the material, while the wave phase at the top areas lagged behind the wave phase at the bottom. Thus, this gradient structure of the material has compelled inner currents to follow curved paths to achieve wave phase matching with each other [52]. Consequently, the gradient impedance design of the FCM-c and FCM-p has altered the current vector orientation of EM waves, which extended the EM wave propagation path and further facilitated the energy dissipation.

Furthermore, the distribution of the electric field intensity and power loss revealed similar variation tendency with the distribution of volume current density in the frequency range of 3.0–15.0 GHz, as illustrated in Fig. S9. Therein, the areas with an intense power loss exactly occurred where the current distribution was concentrated. Moreover, when EM waves were individually radiated at the mid-low frequency peaks of FCM-cp, FCM-cy, and FCM-hp, the EM energy dissipation was mainly localized at the interface between the upper structure and the substrate of units, as well as within the central region of the substrate. In comparison with this, the electric field and associated losses within FCM-c and FCM-p have penetrated more deeply into the center of the material. Besides, when the frequency increased at around 15 GHz, the electric field intensities at the substrate of FCM-cp, FCM-cy, and FCM-hp were all decreased, and the attenuation of EM waves primarily occurred on the upper surface and edges of the super structure, showing a strong edge diffraction effect. Meanwhile, along with the direction of the coordinate Z-axis, the structural design of the cylinders and hexagonal prisms was alike the projection of the cubic prism structure, resulting in that the impedance and absorption performance of FCM-cy and FCM-hp were comparable to those of FCM-cp, as shown in Fig. S9(b and c). It was worth noting, variations in the design of the array unit structure within the xoy plane have a negligible effect on both the absorption bandwidth and the absorption peak [14,26].

On the contrary, as presented in Fig. S9(d and e), when extending from the top of the structure along the negative Z-axis direction to the bottom of the conical or pyramidal structure, both the electric field intensity and power loss intensity inside the FCM-c and FCM-p structures first increased and subsequently decreased, with their maximum values occurring at the junction between the conical/pyramidal structure and the base layer. Hence, this phenomenon indicated that the incident EM waves achieved a deep penetration into the internal structure of FCM-c and FCM-p, rather than being confined to the surface regions. In parallel with that, under high-frequency conditions, electric field concentration was observed at the tip and central regions of the conical and pyramidal structures, as well as in the upper portions of the substrates. As a result, the EM wave at the relatively high frequency band was not reflected by the internal WCNs network, but rather absorbed and dissipated [26]. In summary, the pyramidal geometry of the structure created a desire gradient impedance within the FCM-p, and the absorbing performance for both low- and high-frequency EM waves were enhanced. Thereby, according to the geometrical theory of diffraction, the EM waves would propagate in different forms for distinct boundary properties of diverse structures [50]. Unlike the conical structure of FCM-c, the pyramid structure generated numerous diffracting waves due to its edges, then resulting in electric field concentration accumulating at the apex and edges, thus significantly enhancing EM energy attenuation. However, the smooth transition on the side of the FCM-c weakened this effect, while the absorption capacity of FCM-p was significantly superior to that

of other shaped structures.

3.4.2. Impedance matching analysis

As elaborated above, the EM wave absorption efficiency was mainly enhanced by the geometric gradient configuration of the array unit structure along the EM wave incident direction. Thereby, an explicit correlation was demonstrated in this section between the structural design and the metamaterial's impedance matching, which proved crucial for achieving a broad effective absorption bandwidth of the FCM. The Smith charts were employed to analyze the trends of impedance matching versus frequency for the different FCM structural types, and the results were presented in Fig. S10(a–f). It could be seen that the Smith curves of various FCM were closer to the matching center than those of the foam cement-based material with a simple flat slab structure, indicating the superior impedance matching performance of the metamaterial. Therein, attributing to the structural similarity in their vertical section, the Smith curves of FCM-cy, FCM-cp, and FCM-hp displayed a similar changing trend as the frequency varied. On the other side, the conical and pyramidal array unit structures (FCM-c and FCM-p) have achieved better impedance matching ability owing to their optimized gradient structure design. Therefore, a geometric gradient configuration of the metamaterial has enabled excellent penetration of electromagnetic waves into the metamaterial substrates, then greatly enhancing the overall electromagnetic attenuation performance.

To further elucidate the impedance matching enhancement of the FCM-p, the equivalent impedance Z_i was determined by the equivalent dielectric constant (ϵ_{eq}) and the equivalent permeability (μ_{eq}) of the metamaterial were proposed, depending on the effective medium volume of the material each layer. Thus, the ϵ_{eq} and μ_{eq} of the material could be expressed by the following formulas [2,26]:

$$\epsilon_{eq} = \epsilon_r V + \epsilon_0 (1 - V) \quad (7)$$

$$\mu_{eq} = \mu_r V + \mu_0 (1 - V) \quad (8)$$

Hereinto, ϵ_0 and μ_0 were the air dielectric constant and permeability, V was the volume fraction of the FCM in each layer. According to the transmission line theory [2,46], Z_i and the attenuation constant α_i of the FCM unit layer could be evaluated by the following formulas:

$$Z_i = Z_0 \sqrt{\mu_{eq} / \epsilon_{eq}} \quad (9)$$

$$\alpha_i = \frac{\sqrt{2\pi f}}{c} \sqrt{\sqrt{(\epsilon')^2 + (\epsilon'')^2} - \epsilon'} \quad (10)$$

Since the FCM materials proposed in this study were entirely composed of dielectric materials, no significant magnetic response was observed, then $\mu_r = \mu' = 1$ and $\mu'' = 0$. The normalized attenuation constant A_i could be derived as [26,36]:

$$A_i = \frac{\alpha_i c}{f} = \sqrt{2\pi} \sqrt{\sqrt{(\epsilon')^2 + (\epsilon'')^2} - \epsilon'} \quad (11)$$

Furthermore, the material of FCM was analyzed by dividing the total thickness into 1 mm aliquots, which was based on the effective medium theory [14,26]. When the total thickness was taken as 50 mm, the values of Z_i and A_i per layer for FCM with different biomimetic unit structural types (FCM-cp, FCM-cy, FCM-hp, FCM-c, and FCM-p) were displayed in Fig. S10(g and h). The initial Z_{in} of FCM-cp, FCM-cy, and FCM-hp were 254 Ω , 270 Ω , and 253 Ω , respectively, which presented a distinct deviation from air impedance (377 Ω). Despite their relatively high A_i values (0.52, 0.43, and 0.52, respectively) of FCM-cp, FCM-cy, and FCM-hp, substantial EM waves reflection occurred on the first layer of material. Moreover, with the increase of depth, the Z_{in} values of FCM-cp, FCM-cy, and FCM-hp remained constant continuously. Conversely, on the first layer of array unit FCM-c and FCM-p, the equivalent impedance values were separately 376.6 Ω and 376.7 Ω , which were both rather

closed to 377 Ω . As the depth gradually increased, the effective volume fraction of FCM-c and FCM-p also rose, while the Z_i of each layer progressively decreased. Therefore, an effective coordination and matching between the impedance and the attenuation capability of the FCM-cy and FCM-hp was achieved, which dramatically broadened the –10 dB and –20 dB absorption bandwidths within the 2–18 GHz frequency range. Simultaneously, combining to the effective medium theory and equivalent circuit principle (Eq. (12)) [23,47], the EM wave transmission path in the FCM would be modeled as an equivalent circuit. In this model, the resistance (R), inductance (L), and capacitance (C) were parameters related to the pyramid's height and volume, as well as the array pitch of the pyramid units. Consequently, the impedance of two different kinds of array unit exhibited intrinsic equivalent circuit relationships which was not only with EM parameters of the foam cement-based material but also with its geometric configurations, as displayed in Fig. S10(i and j). The sample of FCM-cp was equivalently composed with two parallel circuits, while the FCM-p sample was divided into multiple cross-sections along the vertical direction, each layer corresponded to an individual RLC loop, therein each FCM-p array unit has yielded into series parallel circuits. With the increase of the structure depth, both the values of C and L were continuously varied, thus the input impedance was changed along the vertical orientation of the pyramid unit.

$$Z_{in} = R + j\omega L + \frac{1}{j\omega C} \quad (12)$$

Moreover, the electric and magnetic field vector distributions at three reflection peaks of 3.60 GHz (S band), 5.76 GHz (C band), and 12.56 GHz (Ku band) were plotted in Fig. S10(k–m). Therein, the frequency point of 3.60 GHz appeared in the lower semicircle of the Smith chart that exhibited capacitive characteristics, and the phase of magnetic field vectors (described by pink arrows) surpassed the electric field vectors (described by blue arrows) at this frequency. Besides, at 5.76 GHz in the upper semicircle, inductive behavior was observed with the electric field leading that of the magnetic field. Meanwhile, the impedance matching ($Z \approx 1$) occurred exclusively when frequency points resided near the central region in Smith chart, as evidenced at 12.56 GHz in Fig. S10(m), where equilibrium between electric and magnetic field vector was established. Noteworthy, when the Smith curve has fallen within the yellow circular area in the Smith chart, it indicated that the RL value of the material at this frequency point was less than –10 dB [46]. Thus, the Smith curve of FCM-p has repeatedly traversed the central zone within the frequency band, which enabling perfect ultra-broadband impedance matching.

3.4.3. The EM wave absorption mechanism analysis of FCM-p

A comprehensive analysis of the interaction between EM waves and FCM-cp, FCM-p, as well as with the air box were compared in Fig. S11. When EM waves were propagating in the air, its electric field vector assumed a sinusoidal spatial distribution with a wavelength determined by its frequency ($\lambda_0 = \frac{c}{f} = \frac{2\pi}{\omega\sqrt{\mu_0\epsilon_0}}$), and experienced negligible attenuation, as shown in Fig. S11(a and d). According to Maxwell's equations [47,53], EM waves experienced a phase shift of π degrees at a propagation distance of $n \cdot \lambda/2$ ($n = 1, 2, 3, \dots$). However, when the incident EM waves entered the FCM, the propagation wavelength inside the metamaterial (λ_m) decreased to $\lambda_m = \frac{\lambda_0}{\sqrt{\epsilon_r}}$ (where λ_0 was the wavelength of light in the air). Since the relative permittivity (ϵ_r') of metamaterial was significantly higher than that of air, the phenomenon of spatial wavelength compression was observed, as shown in Fig. S11(b, e) and Fig. S11(c and f). This wavelength compression effect constituted the physical foundation enabling electromagnetic wave-absorbing metamaterials to achieve broadband absorption through thinner structural layers [11]. Specifically, a reduced propagation distance has triggered more oscillation cycles in the material, thereby significantly enhanced the interference cancellation loss of the EM energy. Furthermore, the

propagation path of the EM wave was marked with black arrows, which was also deemed as the Poynting vectors. At the relatively lower frequency of 3.6 GHz, the electromagnetic wave penetration within the FCM-cp was predominantly localized on the top and substrate surfaces, a distribution attributed to the rectangular geometric design. In comparison, the design of FCM-p permitted continuous wave incidence from pyramidal sides, indicating a greater number of incident channels provided by the gradient design. Nevertheless, as a result of the limitation of skin effect, most EM energy was mainly dissipated near surface region at this frequency for both FCM-cp and FCM-p. When the frequency point was switched at 14.96 GHz, the EM wavelength became commensurate with the either structural dimension of the FCM-cp and FCM-p. Thus, intense EM energy loss was concentrated at the center of the metamaterial, giving rise to a pronounced local resonance and a resonant cavity effect. Additionally, despite a comparable internal propagation wavelengths happened in both FCM-cp and FCM-p, a phase shift occurred at the pyramidal surface of the FCM-p, which has induced an arc-shaped current path that has consequently enhanced the EM wave absorption behavior, as illustrated in Fig. S11(c and f). Moreover, as exhibited in Fig. S11(g), the electric field intensity and vector distributions among the FCM-p array structural units were also analyzed at the same frequency point of 14.96 GHz. It was found in Fig. S11(g2), in the xoy plane, strong electric fields predominantly occurred along four vertical edges of each pyramid unit, in the surroundings at the mid-height of each unit, and in the adjacent area among different bases of pyramid units. Alternatively, in the yoz cross-section, as shown in Fig. S11(g3), the Poynting vectors were bent at the pyramid surface and then entered the FCM-p through the side facets. Eventually, the Poynting vectors converged toward the center of the unit structure, which was accorded well with the electric field vector distribution observed in the xoy plane. Such bending behavior of the EM wave propagation path can be attributed to two factors: the refraction of the wave at the interfaces between different media and multiple reflections within the gaps among

the units [52]. As the EM wave deeply propagated inside the FCM-p, multiple intense electric fields were also induced along the direction of wave incidence. At the same time, based on the $\lambda/4$ theory (Eq. (4)), interference cancellation across varying thicknesses was achieved, and this achievement was ascribed to the pyramid structural array unit's design. Therefore, the vector superposition of all incident and scattered EM waves inside the FCM-p has ultimately led to the phenomenon of EM energy loss converging toward the center of the cone.

In summary, the EM wave absorption and defense mechanisms of FCM-p metamaterials was detailed in Fig. 8. Owing to sources of EM radiation, which mainly include natural radiation caused by natural phenomena (such as thunderstorms and tornadoes), non-ionizing radiation from radio waves, microwaves, and similar frequencies, as well as active radiation generated by artificial devices, human activities (such as televisions, computers, and mobile phones) and military detection technology (such as reconnaissance satellite, missile, and unmanned aerial vehicle), EM waves were ubiquitously present in the spaces of human life and industrial production [54]. Thereby, the uncontrolled EM radiation not only disrupted daily life and the normal functioning of medical equipment and other precision instruments, but also posed a threat to military and national defense security. The FCM, enlightened by the bionic structural designs, was fabricated to offer superior architectural adaptability and durability, making it suitable for application on the walls of existing buildings upgrading, where it can provide electromagnetic silence, shielding or enhance thermal insulation and energy efficiency within building spaces. To that end, an FCM-p with bionic structural pyramid array units inspired by *Apriona germari* wing hairs was designed with an optimized multiscale structure to enhance EM wave absorption efficiency over a broader frequency range, as well as to provide insensitivity to different angles and modes of EM wave incidence. On the first level, when the EM wave reached the surface of FCM-p, the pyramidal design of the array unit structure provided a gradient impedance profile, facilitating the penetration of more EM

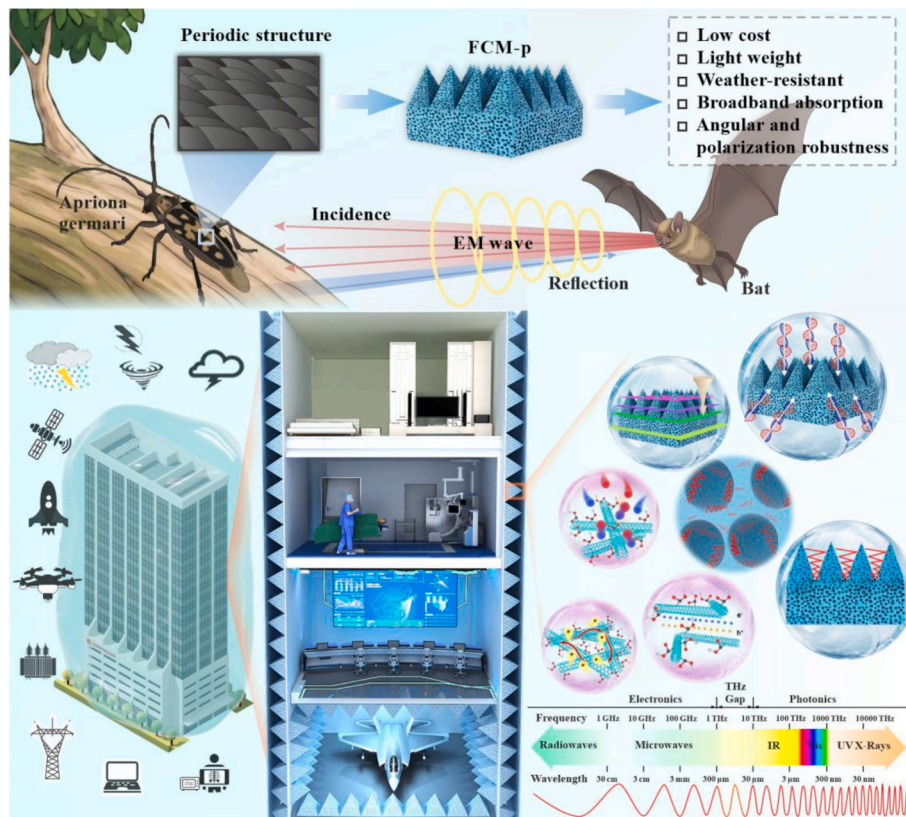


Fig. 8. The mechanisms of FCM-p incorporated with MWCNTs.

waves into the interior of the material. Alternatively, different from traditional EM absorbing materials, the metasurface of FCM regulated the EM spectrum through the electric and magnetic resonance modes excited in its macro- and microstructural units. By ingeniously adjusting the geometric shape, size, and arrangement of the structural array units, the resonant behavior of the EM modes can be flexibly tailored, thus enabling control over the EM spectral response [5]. As a result, the serious EM resonance happened between EM waves and the periodic array unit structures, which dramatically increased the effect of multiple reflection, interference cancellation, and scattering attenuation of the EM waves. On the other side, due to the characteristics of the porous structure of the foaming cementitious matrix, the propagation path of the EM waves within the FCM-p metamaterial matrix was prolonged, which further enhanced the multiple reflection and refraction of EM wave in the pores, thus the wave absorbing performance and efficacy of the FCM-p was remarkably improved. Simultaneously, on the micro-scale level, the incorporation of MWCNTs has possessed high conductivity and excellent EM dielectric loss performance, into the cement-based foam metamaterial introduced additional loss mechanisms from a microstructural perspective [21,55,56]. These included interfacial polarization, ionic polarization, and electric dipole polarization resulting from the defects and functional groups present on the MWCNTs, along with the extensive interfacial polarization loss occurring between the MWCNTs, cement matrix, and air pockets. Thus, the above mechanisms collectively contributed to further dissipation of electromagnetic wave energy. Moreover, the uniformly dispersed MWCNTs within the matrix formed a three-dimensional conducting network, further facilitated the migration and leap paths of excited state electrons under alternating EM fields, and improving conductive losses. Therefore, a high efficiency and broadband EM waves absorption by combining perfect impedance matching and multiscale EM waves loss mechanisms was achieved through the bionic design of FCM-p which enlightened by the array structure of *Apriona germari* wing hairs.

4. Conclusions

Based on the guiding ideology of EM absorbing metamaterial structures, FCMs inspired by different biological structures such as chameleon skin, moth eyes, proteinaceous spherules on leafhopper cuticles, crocodile skin, and *Apriona germari* wing hairs, corresponding to five geometrically distinct periodic-array FCM absorbers were respectively designed and investigated in this work. The EM wave absorption performances of the FCM with five different geometry structures with various dimensional parameters, and incidence angles and the relationship between the macrostructure and the wave absorbing properties of FCM were comprehensively compared and analyzed by the computer simulation software HFSS. From the results, the FCM-p metamaterial inspired by the *Apriona germari* wing hairs has demonstrated ultra-broadband and high absorption capability. It has displayed effective EM wave absorption in the range of 3.36–18 GHz. Moreover, the qualified EM absorption of $RL < -20$ dB was achieved for X and Ku bands, and the $RL_{\min}$ of -63.53 dB was obtained at 9.53 GHz. In addition, when the incident angle was increased from 0° to 60° , excellent EM wave absorbing characterization of the FCM-p was still maintained. Therefore, the design of the pyramid array structure endowed the FCM-p with excellent gradient impedance. Such structure then interacted with multiple loss mechanisms (such as interference cancellation, interfacial polarization, and conductive loss) within the substrate, leading to the excellent EM absorption performance of the FCM-p, which showed great potential applications in the field of EM protection in the future smart buildings.

CRediT authorship contribution statement

Xianyun Shi: Writing – original draft, Software, Investigation, Data curation. **Gan Deng:** Writing – original draft, Software, Investigation,

Data curation. **Wanxia Fang:** Software, Investigation. **Zhaoheng Wang:** Software, Data curation. **Liangjun Yin:** Visualization, Resources, Funding acquisition. **Tingting Yang:** Project administration, Writing – review & editing, Visualization, Funding acquisition. **Yi Deng:** Writing – review & editing, Funding acquisition. **Yuanyi Yang:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

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